

Notes: Berry (Econometrica, 1992)

Estimation of a Model of Entry in the Airline Industry

Contents

1	Big picture	3
1.1	Incremental contribution of the paper	3
2	Data construction and measurement	4
2.1	Market definition: city-pair airline markets	4
2.2	Airline service and the 90-passenger threshold	4
2.3	Timing: entry and exit between two quarters in 1980	4
2.4	Market-level variables	5
2.5	Firm-level airport-presence variables	5
3	Model and empirical specifications	5
3.1	Entry game and equilibrium condition	5
3.2	Profit specification with strategic interaction and firm heterogeneity	6
3.3	Unique equilibrium number of firms	6
3.4	Special-case estimators	6
3.5	Simulation estimator	7
4	Identification and empirical strategy	7
4.1	What identifies airport-presence effects	7
4.2	Why the model must include strategic interaction	7
4.3	Why firm heterogeneity matters	8
4.4	Role of order-of-entry assumptions	8
4.5	Threats to identification	8
5	Tables and figures	9
5.1	Table I: The Joint Frequency Distribution of Entry and Exit + Table II: Number and Percentage of Markets Entered and Exited in the Large City Sample, by Airline	9
5.2	Table III: Number of Potential Entrants, by Number of Cities Served within a City Pair + Table IV: Regression Results for Number of Firms	9
5.3	Table V: Probit Results + Table VI: Maximum Likelihood Results	10
5.4	Table VII: Simulation Estimates + Table VIII: Predictions from the Models	11
6	Interpretive synthesis	12
6.1	Why airport presence matters	12
6.2	Why entry alone is informative but incomplete	12
6.3	Why the simulation estimator matters	12
6.4	How the paper should be read today	13

7	Short summary	13
8	Conclusion	13
8.1	Core conclusion	13
8.2	Incremental contribution revisited	13
8.3	Practical implication	14
8.4	What the paper does not show	14
8.5	Takeaway	14

1 Big picture

- **Question.** How much does an airline’s airport presence raise the profitability of serving a city-pair route, and how should one estimate such a profit effect when route entry is an oligopoly game with many heterogeneous potential entrants?
- **Main empirical setting.** The paper studies entry and exit in U.S. airline city-pair markets shortly after deregulation. A market is passenger travel between two cities, not necessarily nonstop service.
- **Core data source.** The analysis uses the U.S. Origin and Destination Survey of Air Passenger Traffic, especially Data Bank 4, which records passenger counts by airline and city pair on an origin-and-destination basis.
- **Main economic idea.** Entry choices reveal underlying route profitability. A firm enters a market if expected post-entry profits are positive, taking into account market size, distance, its own airport presence, and expected rivals.
- **Strategic complication.** Entry is not a collection of independent binary decisions. Profitability depends on the number of entrants because more firms in the city pair reduce profit through competition.
- **Heterogeneity complication.** Airlines differ in airport presence, route networks, costs, fleets, labor contracts, and management. These differences matter for who enters, not just for how many firms enter.
- **Computational contribution.** With many potential entrants, the probability region associated with an observed equilibrium can be extremely complex. The paper uses simulation estimators to avoid evaluating infeasible high-dimensional regions exactly.
- **Main result on airport presence.** Prior service at both endpoint cities is strongly associated with entry into the city-pair market. In the descriptive table, firms that already serve both cities enter at 6.46%, compared with 0.36% for firms serving only one city and 0.01% for firms serving neither.
- **Main structural result.** Airport presence raises profitability, but profits decline substantially with the number of entering firms. Thus, policies that improve airport access increase entry less than a simple probit model would predict.
- **Policy implication.** Reducing airport entry barriers may not mechanically generate large numbers of new competitors because within-market competition limits the number of firms that can profitably operate.

1.1 Incremental contribution of the paper

- **Relative to discrete-choice entry models.** Earlier binary-choice models use entry as a proxy for latent profitability, but Berry embeds this idea in an oligopoly entry equilibrium. This is crucial because the action of one airline changes the payoffs of other airlines.
- **Relative to Bresnahan-Reiss style entry models.** Bresnahan and Reiss study market size and the equilibrium number of firms, often abstracting from rich firm-specific heterogeneity. Berry’s paper focuses on many heterogeneous potential entrants and asks which firms enter, not only how many enter.
- **Relative to airline hubbing studies.** Prior work documented hub premiums and airport-presence advantages in prices or fares. This paper estimates the profit advantage of airport presence from entry choices and uses the estimates to simulate changes in market structure.
- **Relative to simulation-estimator applications.** The paper gives a new reason to use simulation estimators: not merely aggregation or simple intractable integrals, but the complicated integration regions generated by equilibrium conditions in discrete games with many heterogeneous firms.
- **Conceptual contribution.** The paper makes clear that policy counterfactuals can be misleading if one ignores either strategic interaction or firm heterogeneity. A probit model can fit observed entry reasonably well but greatly overstate the response to airport-access policies.

Interpretation: The incremental contribution is methodological and substantive. Methodologically, the paper demonstrates how simulated method-of-moments ideas can make equilibrium entry models estimable when exact likelihoods are infeasible. Substantively, it shows that airport presence is valuable but that competition constrains the total number of entrants.

2 Data construction and measurement

2.1 Market definition: city-pair airline markets

- The unit of observation is a city-pair market, such as travel between city A and city B .
- The market is defined on an origin-and-destination basis. A passenger can travel through a hub; the service need not be nonstop.
- The sample is restricted to city pairs connecting the 50 largest U.S. cities, giving 1,225 possible city-pair combinations.
- Six city pairs do not appear in the O&D survey and are omitted, leaving 1,219 markets in the main structural analysis.
- Multiple airports in a city are aggregated because the O&D data identify city-level origin-destination flows rather than airport-level products.

Interpretation: The city-pair definition matches how passengers choose an origin and final destination, but it means that entry can occur through connecting service. This matters because hub-and-spoke networks can generate route entry without nonstop route creation.

2.2 Airline service and the 90-passenger threshold

- The O&D survey is a 10% random sample of airline tickets.
- A firm is counted as serving a city-pair market if at least 90 passengers appear in the 10% O&D sample.
- This threshold corresponds roughly to one weekly flight with a medium-size jet and avoids treating coding errors or very indirect itineraries as active market service.
- Berry reports that special-case estimates are not very sensitive to changes in this threshold.

Interpretation: The threshold translates a continuous passenger count into a discrete entry indicator. It is important because the entire model treats market participation as a binary choice.

2.3 Timing: entry and exit between two quarters in 1980

- The first period is the first quarter of 1980.
- The second period is the third quarter of 1980.
- New entry occurs when a firm does not serve a city pair in the first period but serves it in the second period.
- Exit occurs when a firm serves a city pair in the first period but not in the second.
- The six-month window is intended to be long enough for entry decisions but short enough that demand and cost fundamentals are relatively stable.

Interpretation: The two-quarter design makes the paper closer to a short-run entry analysis than a long-run network-dynamics model. It captures route-level adjustment during the early deregulation period, before later merger waves and the air traffic controllers strike.

2.4 Market-level variables

- **Population.** Berry uses the product of the populations of the two endpoint cities, measured in tens of millions. This proxies for potential demand in a gravity-style way.
- **Distance.** Distance between the two cities is measured in thousands of miles. Distance can affect both demand and costs, so its sign is not necessarily obvious.
- **Distance squared.** Some descriptive and probit specifications allow nonlinear distance effects.
- **Tourist market dummy.** A tourist dummy equals one for markets in Florida and California and zero otherwise.
- **Number of firms.** The equilibrium number of firms in the second period is a central outcome and also a determinant of profitability in the model.

Interpretation: Market size raises the capacity of a route to support firms. Distance and tourist status proxy for route economics. However, the central empirical challenge is that the number of firms is simultaneously an outcome and a determinant of profits.

2.5 Firm-level airport-presence variables

- **City2.** A dummy equal to one when a potential entrant serves both endpoint cities in the first period.
- **City N2.** The number of potential entrants that serve both endpoint cities.
- **City N+.** The number of potential entrants that serve only one endpoint city.
- **City share.** A continuous measure of the firm's average airport output share across the two endpoint cities, measured in passenger-miles.
- **Number of routes.** An alternative airport-presence measure, equal to the mean number of destinations served out of the origin and destination cities.

Interpretation: These variables operationalize the idea that airline networks and airport presence create profitability advantages. The binary City2 variable captures broad access to both endpoints, while City share captures intensity of presence.

3 Model and empirical specifications

3.1 Entry game and equilibrium condition

In each market i , there are K_i potential firms. Firm k chooses $s_{ik} \in \{0, 1\}$, where 1 means operating in the market and 0 means not operating. Let s_i be the vector of entry decisions and $N(s_i)$ be the number of entering firms.

The equilibrium condition is that all entering firms earn nonnegative profits and all non-entering firms would not profit from unilateral entry:

$$s_{ik}\pi_{ik}(s_i^*) > 0, \quad (1 - s_{ik}^*)\pi_{ik}(s_i^{*+k}) \leq 0, \quad k = 1, \dots, K_i.$$

Interpretation: This formalizes why entry cannot be treated as an isolated binary decision. A firm's profitability depends on who else is in the market and how many competitors it faces.

3.2 Profit specification with strategic interaction and firm heterogeneity

Berry imposes a profit form:

$$\pi_{ik}(s_i) = v_i(N(s_i)) + \phi_{ik},$$

where $v_i(N)$ is common variable profit that declines in N and ϕ_{ik} is a firm-market profitability index.

The empirical form is:

$$\pi_{ik}(N) = X_i\beta + Z_{ik}\alpha - \delta \ln(N) + u_i + u_{ik},$$

where:

- X_i are market characteristics such as population and distance;
- Z_{ik} are firm-market characteristics such as airport presence;
- δ measures how profits decline with the number of entrants;
- u_i is market-level unobserved profitability;
- u_{ik} is firm-market unobserved profitability.

Interpretation: The coefficient δ is the strategic-competition term. It is the key distinction between the full entry model and a probit model that treats each airline's entry decision as independent.

3.3 Unique equilibrium number of firms

Given the profit form above and strictly decreasing $v_i(N)$, all pure-strategy equilibria involve the same number of firms, even if the identities of entering firms may not be unique.

Order firms by their profitability index, $\phi_{i1} \geq \phi_{i2} \geq \dots \geq \phi_{iK_i}$. The equilibrium number is:

$$N_i^* = \max\{n : v_i(n) + \phi_{in} > 0\}.$$

Interpretation: This is the core tractability result. The identities of entrants may be ambiguous, but the number of entrants can be pinned down, which makes estimation feasible using the observed number of firms.

3.4 Special-case estimators

Berry compares the full model to simpler special cases:

- **OLS count regression.** Regresses the observed number of firms on market characteristics and counts of potential entrants.
- **Simple probit.** Models individual entry as if firm profits are unaffected by the number of entrants.
- **Ordered probit/no heterogeneity model.** Uses the number of entrants but abstracts from observed firm heterogeneity.
- **Special maximum likelihood cases.** Restrict the error structure enough to make likelihood evaluation possible.

Interpretation: These models are not merely robustness checks. They reveal why ignoring strategic interactions can produce misleading policy counterfactuals even when within-sample fit looks acceptable.

3.5 Simulation estimator

The full likelihood is computationally hard because the probability of an equilibrium outcome involves complicated regions of integration. Berry rewrites estimation as a method-of-moments problem.

Define the prediction error for the number of firms:

$$v_{i0} = N_i^* - E[N_i^* \mid W_i, \theta],$$

where W_i collects market and firm characteristics. At the true parameter vector,

$$E[v_{i0} \mid W_i, \theta_0] = 0.$$

The difficult conditional expectation is replaced by a simulated unbiased estimator. For a simulated draw of errors, define:

$$\hat{n}(W_i, \theta, u_i) = \max\{n : \#\{k : \hat{\pi}_{ik}(n, \theta, u_i) > 0\} \geq n\}.$$

With T simulation draws:

$$\hat{N}(W_i, \theta) = \frac{1}{T} \sum_{t=1}^T \hat{n}(W_i, \theta, u_i^t).$$

The estimator minimizes a quadratic form in sample moments:

$$G(\theta) = g(\theta)' A g(\theta).$$

Interpretation: Simulation does not simplify the economics. It approximates the conditional expectation implied by the equilibrium model. This lets the model keep firm heterogeneity and strategic interaction without evaluating a massive exact likelihood.

4 Identification and empirical strategy

4.1 What identifies airport-presence effects

The airport-presence coefficients are identified by variation across firms within the same city-pair market and across markets with different endpoint service patterns. In particular, a firm can have no endpoint presence, one-city presence, or two-city presence, and the paper asks whether these differences predict entry conditional on market-level demand and cost variables.

Interpretation: The identification is not experimental. It is structural and cross-sectional: entry decisions are used as revealed-profitability data, and airport-presence variables proxy for firm-specific profitability advantages.

4.2 Why the model must include strategic interaction

If airport access improves for many firms, the profitability of each firm also changes because more firms may enter. A simple probit captures the direct effect of airport presence but ignores the crowding-out effect of additional entrants on profit. The structural model separates these channels by including $-\delta \ln(N)$ in the profit equation.

Interpretation: The key identification logic is that policy counterfactuals require a model of equilibrium. The direct effect of being a better potential entrant is not enough; one must also know how entry changes competitive conditions.

4.3 Why firm heterogeneity matters

Heterogeneity matters because the identity of potential entrants changes with airport presence. Some airlines are naturally better positioned for some city pairs because of endpoint service, route networks, or hub structure. If the model assumes symmetric firms, the relationship between market size and number of firms can be misinterpreted as evidence of competitive effects rather than heterogeneity in suitability.

Interpretation: Berry’s model shows that both forces can explain more firms in larger markets. Identification requires variables that shift the set of potential entrants or their profitability, not only market size.

4.4 Role of order-of-entry assumptions

Because the identities of entrants can be nonunique, Berry considers order assumptions: most profitable firms move first, or incumbents move first. The latter is motivated by the possibility that incumbents or hub carriers control airport bureaucracies or have first-mover advantages.

Interpretation: Order assumptions are not just technical details. They determine how the model maps latent profits into entrant identities, which is why the paper reports both the most-profitable-move-first and incumbents-move-first simulation estimates.

4.5 Threats to identification

- **Endogenous airport presence.** Airlines may build airport presence in cities where routes are inherently profitable. This means airport presence may partly proxy for unobserved demand or cost advantages.
- **Entry threshold.** The 90-passenger cutoff may misclassify thin service, though the paper notes that special-case estimates are not very sensitive to the cutoff.
- **Partial-equilibrium assumption.** The model treats each city pair separately rather than modeling the entire airline network.
- **No price data.** Since Data Bank 4 lacks prices, the model cannot decompose profitability into fare, cost, and quantity channels.
- **Welfare limits.** Entry data alone cannot show consumer surplus, price competition intensity, or whether potential entry constrains prices.

Interpretation: The paper is best read as estimating the profitability implications of airport presence and demonstrating an estimation method for entry games, not as a complete welfare analysis of airline deregulation.

5 Tables and figures

5.1 Table I: The Joint Frequency Distribution of Entry and Exit + Table II: Number and Percentage of Markets Entered and Exited in the Large City Sample, by Airline

Read:

- Table I shows that new entry and exit are both common: new entry occurs in more than 20% of markets and exit in about 14% of markets.
- The modal outcome is no entry and no exit: 68.50% of markets are stable over the six-month window.
- Table II shows large differences across airlines in entry and exit behavior.
- Ozark enters 64.3% of markets served in the first period, while Pan Am enters only 2.2%.
- Berry notes that the across-firm correlation between entry and exit rates is strongly positive at 0.695, consistent with network reallocation rather than a simple expansion-only story.

Interpretation: Table I establishes that entry and exit are frequent enough to make entry choices informative. Table II shows that airlines are not homogeneous entities reacting identically to market size. The same firms can enter some markets and exit others, suggesting that the relevant heterogeneity is market-specific suitability, not merely a firm-level fixed effect.

Economic message: The descriptive evidence motivates the structural model: a route-level entry model must allow both churn and heterogeneous airline-market fit.

Number of Exits, as % of Total Markets in the Sample:						
	0	1	2	3 +	Total	
Number	0	68.50	10.01	1.07	0.00	79.57
of	1	15.09	2.63	0.41	0.00	18.13
Entrants	2	1.96	0.25	0.00	0.00	2.05
(as %)	3 +	0.16	0.08	0.00	0.00	0.24
	Total	85.56	12.96	1.48	0.00	100.00

(a) Table I: Joint frequency distribution of entry and exit

TABLE II
NUMBER AND PERCENTAGE OF MARKETS ENTERED AND EXITED IN THE LARGE CITY SAMPLE, BY AIRLINE

	Airline	# of Markets Served	# of Markets Entered	# of Markets Exited	% of Markets Entered	% of Markets Exited
1	Delta	281	43	28	15.3	10.0
2	Eastern	257	33	36	12.8	14.0
3	United	231	36	10	15.6	4.3
4	American	207	22	12	10.6	5.8
5	USAir	201	20	17	10.0	8.5
6	TWA	174	22	23	12.6	13.2
7	Braniff	112	10	20	8.9	17.9
8	Northwest	75	6	7	8.0	9.3
9	Republic	69	9	6	13.0	8.7
10	Continental	62	9	5	14.5	8.1
11	Piedmont	61	14	2	23.0	3.3
12	Western	51	6	7	11.8	13.7
13	Pan Am	45	1	1	2.2	2.2
14	Ozark	28	18	4	64.3	14.3
15	Texas Int'l	27	3	6	11.1	22.2

(b) Table II: Entry and exit by airline

5.2 Table III: Number of Potential Entrants, by Number of Cities Served within a City Pair + Table IV: Regression Results for Number of Firms

Read:

- Table III shows a striking airport-presence gradient: entry is 0.01% for firms serving neither endpoint city, 0.36% for firms serving one endpoint city, and 6.46% for firms serving both endpoint cities.
- The large increase from one-city to two-city presence suggests that endpoint access is not just a generic scale variable; serving both endpoints creates a qualitatively different level of readiness to enter.
- Table IV shows that the number of firms is strongly related to market size and to the number of potential entrants with endpoint presence.

- The coefficient on City $N2$ is 0.338, meaning an additional potential entrant serving both endpoint cities is associated with about one-third of an additional firm in the market.

Interpretation: These tables are the empirical bridge from descriptive facts to the structural model. Table III shows that firm identity and endpoint presence matter for individual entry. Table IV shows that market structure also depends on the pool of qualified potential entrants.

Economic message: Airport presence appears to be economically meaningful, but the count regression cannot distinguish whether more firms appear because profits are higher, because firms are more suitable, or because strategic interaction is weak. The structural model is needed for that separation.

TABLE III
NUMBER OF POTENTIAL ENTRANTS, BY NUMBER OF CITIES SERVED WITHIN
A CITY PAIR, WITH NUMBER AND PERCENTAGE ENTERING

Number of Cities Served	Total # of Potential Entrants	# Entering	% Entering
0	47600	4	0.01%
1	12650	45	0.36
2	3590	232	6.46

(a) Table III: Potential entrants and airport presence

TABLE IV
REGRESSION RESULTS FOR NUMBER OF FIRMS

Var	Est Parm (Std. Error)	Mean Value of Var. (Std. Dev.)
N		1.629 (1.393)
Const	-0.727 (0.097)	—
Pop	2.729 (0.255)	0.558 (0.114)
Dist	-1.591 (0.827)	1.149 (0.093)
Dist ²	0.337 (1.850)	0.022 (0.039)
Tourist	0.134 (0.089)	0.116 (0.320)
City $N2$	0.338 (0.011)	4.574 (2.684)
City $N +$	0.084 (0.009)	10.377 (3.656)
R -squared is: 0.612		

(b) Table IV: OLS count regression

5.3 Table V: Probit Results + Table VI: Maximum Likelihood Results

Read:

- Table V treats firm entry as an individual probit choice. City2 and City share are large and significant, reinforcing the airport-presence result.
- In Table V column (1), City2 has coefficient 2.05 and City share has coefficient 5.58; these are strong associations with entry.
- Table VI estimates special maximum likelihood cases. The no-heterogeneity model finds that profits decline with N , with $\delta = 1.81$.
- The only-observed-heterogeneity model still finds positive airport-presence effects, but the paper argues that it imposes strong restrictions on entrant identities.
- The no-correlation model gives puzzling signs for city share and a small competition effect, illustrating sensitivity to restrictive error assumptions.

Interpretation: Table V is useful but incomplete because it ignores strategic interaction. Table VI shows that strategic competition and heterogeneity are both necessary. The restrictive models can fit pieces of the data, but they either cannot explain firm identities or imply counterintuitive relationships.

Economic message: The tables justify moving beyond simple probit and ordered probit. In entry games, good policy prediction requires modeling both who is advantaged and how profitability falls as more firms enter.

TABLE V
PROBIT RESULTS^a

Variable:	(1)	(2)	(3)
Const	-3.53 (0.069)	-3.23 (0.101)	-3.36 (0.064)
Pop	1.21 (0.082)	0.18 (0.100)	1.21 (0.082)
Dist	0.03 (0.004)	0.010 (0.005)	1.18 (0.169)
Dist ²	-0.001 (0.0001)	-0.001 (0.001)	—
Tourist	0.12 (0.044)	-0.071 (0.050)	—
City2	2.05 (0.054)	1.83 (0.054)	2.06 (0.054)
City share	5.58 (0.164)	—	5.47 (0.162)
# Routes	—	0.732 (0.018)	—
-2 log-likelihood:	6799	6220	6834

^aObservations are 18218 firm/market combinations. Standard errors are in parentheses.

(a) Table V: Probit results

TABLE VI
MAXIMUM LIKELIHOOD RESULTS^a

Variable	No Heterogeneity	Only Observed Heterogeneity	No Correlation
Constant	1.00 (0.056)	-0.973 (0.485)	-1.54 (0.815)
Population	4.33 (0.102)	4.16 (0.180)	4.32 (0.059)
Dist	-0.184 (0.034)	-0.841 (0.070)	-0.903 (0.112)
City2	—	1.68 (0.479)	1.43 (0.524)
City share	—	1.20 (0.118)	-2.94 (0.070)
δ	1.81 (0.050)	1.66 (0.470)	0.252 (1.92)
-2 log-likelihood:	3715	3619	1732

^aObservations are 1219 markets. Standard errors are in parentheses.

(b) Table VI: Maximum likelihood special cases

5.4 Table VII: Simulation Estimates + Table VIII: Predictions from the Models

Read:

- Table VII reports simulation estimates for two entry-order assumptions: most profitable firms move first and incumbents move first.
- In both columns, City2 and City Share remain economically important. The incumbents-move-first model estimates City2 at 5.91 and City Share at 5.41.
- The estimated competition effect differs sharply across assumptions. In the incumbents-move-first model, $\delta = 4.90$, implying rapid profit decline as the number of firms rises.
- Table VIII compares predictions from probit, ordered probit, and the full model. The full model predicts the actual mean number of firms more closely than probit or ordered probit.
- The policy simulations show why strategic interaction matters. If all firms are made “well-qualified,” probit predicts 6.334 firms on average, while the full model predicts only 2.080.
- The full model also predicts 92.75% correctly classified entrants, compared with 90.29% for the probit.

Interpretation: Table VII is the paper’s central structural result: airport presence matters, but competition also matters. Table VIII shows that the main gain from the structural model is not necessarily a huge increase in within-sample fit. The gain is in counterfactual discipline. A model without strategic interaction dramatically overstates how much airport-access policies would increase entry.

Economic message: Structural entry models are valuable because policy experiments change equilibrium conditions. The probit model can look plausible in sample but fail when used to predict entry after a policy that changes many firms’ access simultaneously.

TABLE VII
SIMULATION ESTIMATES^a

Variable	Most Profitable Move First	Incumbents Move First
Constant	− 5.32 (0.354)	− 3.20 (0.258)
Population	1.36 (0.239)	5.28 (0.343)
Dist	1.72 (0.265)	− 1.45 (0.401)
City2	4.89 (0.295)	5.91 (0.149)
City Share	4.73 (0.449)	5.41 (0.206)
δ	0.527 (0.119)	4.90 (0.206)
ρ	0.802 (0.105)	0.050 (0.048)
Value of the objective fn:	33.3	26.2

^aObservations are 1219 markets. Standard errors are in parentheses.

(a) Table VII: Simulation estimates

TABLE VIII
PREDICTIONS FROM THE MODELS
ACTUAL MEAN NUMBER OF FIRMS: 1.629

		Predicted Mean Number of Firms No Heterogeneity (Ordered Probit)		
		Probit	Full Model	
1.	Actual exog. data	1.326	1.887	1.699
2.	.5 Mil. popl. increase	10.78	2.308	1.809
3.	500 extra miles dist.	0.6710	1.770	1.406
4.	All entrants “Well-Qual”	2.724	—	2.045
5.	All firms “Well-Qual”	6.334	—	2.080
6.	10% incr. in City Shr	2.178	—	1.697
% Correctly Predicted Entrants				
7.		90.29	—	92.75
Mean Squared Error in the Prediction of N :				
8.		1.257	2.072	1.299

(b) Table VIII: Model predictions and policy experiments

6 Interpretive synthesis

6.1 Why airport presence matters

Airport presence can matter through several channels: better gate access, existing local operations, customer loyalty, frequent-flyer programs, hub scheduling, and cost economies from connecting traffic. Berry does not identify which mechanism dominates. Instead, he estimates the reduced-form profitability premium associated with airport presence within an equilibrium entry model.

6.2 Why entry alone is informative but incomplete

Entry is a useful revealed-profitability indicator because firms generally operate only when expected profits are positive. However, entry alone cannot identify the full welfare effect of competition. A market with two firms could be nearly competitive if price competition is fierce, or it could still involve substantial market power. Berry therefore uses entry to infer profitability and policy-relevant market structure, not consumer surplus.

6.3 Why the simulation estimator matters

The simulation estimator makes an otherwise infeasible model operational. The challenge is not merely high dimensionality. It is the complicated shape of the equilibrium region implied by the game. Simulation replaces exact integration over that region with simulated predictions and moment conditions.

6.4 How the paper should be read today

The paper is an early and influential example of estimating empirical games. It anticipates later work in industrial organization that uses equilibrium restrictions, simulation, and structural counterfactuals. The airline context is especially useful because firm heterogeneity is observable and economically important.

7 Short summary

- The paper estimates an oligopoly entry model for U.S. airline city-pair markets.
- Entry decisions are treated as indicators of latent profitability.
- Airport presence is measured by whether an airline serves one or both endpoint cities and by continuous city output shares.
- The data come from the O&D survey for first and third quarters of 1980.
- Markets are city pairs among the 50 largest U.S. cities, yielding 1,219 usable markets.
- Descriptive tables show substantial entry and exit and strong variation across airlines.
- Firms serving both endpoint cities are much more likely to enter a city-pair market.
- The formal model allows profits to depend on market variables, firm-market airport presence, and the number of entering firms.
- A key restriction makes the equilibrium number of entrants unique, even if entrant identities may be nonunique.
- Special-case models illustrate the limitations of simple probit and ordered probit approaches.
- Simulation estimation handles the complex integration regions generated by a many-firm entry game.
- The full model finds that airport presence raises profitability and that profits decline rapidly with entry.
- Counterfactuals show that simple probit models overstate the entry effects of policies that improve airport access.
- The paper cannot conduct welfare analysis because it observes entry, not prices or consumer surplus.

8 Conclusion

8.1 Core conclusion

Berry shows that airline airport presence is strongly related to profitability and entry in city-pair markets, but the effect of airport access on market structure is moderated by strategic interaction. More firms may become qualified to enter, yet not all can profitably operate once competition increases.

8.2 Incremental contribution revisited

The paper's lasting contribution is to transform entry data into structural evidence about profit determinants in an oligopoly game with many heterogeneous firms. It moves beyond the idea that market size alone explains entry and shows how firm-specific advantages and competitive externalities jointly determine observed market structure.

8.3 Practical implication

For policy, the paper warns against extrapolating from individual entry propensities to market-level outcomes. Improving airport access for one airline may raise its entry probability substantially. Improving airport access for many airlines at once also increases competition, which limits equilibrium entry.

8.4 What the paper does not show

The paper does not identify the welfare effect of additional entry, does not separate airport-presence mechanisms, and does not model full airline network dynamics. It should therefore be viewed as a structural entry and profitability study rather than a complete evaluation of deregulation or airport-access policy.

8.5 Takeaway

The central takeaway is that firm heterogeneity and strategic interaction must be modeled together. In airline entry, airport presence makes some firms better positioned to enter, but the market's ability to support entrants is constrained by competition. Simulation methods make this insight empirically usable.